Polynomial and Laurent hulls at a Diaz candidate, and a barrier for generic candidates

Research note for the `diaz-modulus-lean` project
(written by Claude, an AI model, at the request of Carlo Perassi)

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Abstract

Diaz's conjecture says that a non-zero logarithm of an algebraic number never has algebraic modulus. A hypothetical counterexample u, a *candidate*, certifies the three-dimensional space $H_0 = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$ inside $\widetilde{\mathcal{L}}$, the $\overline{\mathbb{Q}}$-span of 1 and the logarithms. Roy's strong six exponentials theorem then excludes a number z from $\widetilde{\mathcal{L}}$ whenever the conjugation hull $W_z = H_0 + \overline{\mathbb{Q}}z + \overline{\mathbb{Q}}\bar{z}$ carries a 2×3 configuration. We decide this for every polynomial expression z in u and $\bar{u}$.

Theorem C (polynomials in u): $W_{P(u)}$ carries a configuration if and only if $\deg P \in \{2, 3\}$. The "if" half is one substitution in Diaz's Théorème 7(1) of 2007, which the project has formalised. What was not found in the sources read is the "only if" half: Roy's theorem cannot exclude from $\widetilde{\mathcal{L}}$, through its own hull, any polynomial in u of degree at least 4. The proof normalises a configuration to Laurent polynomials. The orders at 0 and at ∞ then pinch the supports of the products into $[-1, 1]$.

Theorem E (polynomials in u and $\bar{u}$) classifies the hulls that carry a configuration into three classes. Class (P) is Theorem C. Class (Q) is a family in the exponents $\pm 2, \pm 3$, such as $\mathrm{Re}(u^3) + cu^2$. Its configurations have a factor whose involution exchanges 0 and ∞. They are configurations of Diaz's Théorème 2 of 2004, and the family is the cyclic family of the project's research notes for that involution, which the notes only sketch. Class (R) is the project's real pencil spaces. In particular, when the exponents stay within ± 3, a three-factor equation in the coefficients decides the question. When some exponent has absolute value at least 4, only class (R) remains. Real polynomial expressions in $\mathrm{Re}\, u$ and $\mathrm{Im}\, u$ with algebraic coefficients are never excluded.

Theorem D is a barrier. Adjoin to the logarithms a point u of a circle of algebraic radius that is generic over the field they generate. The enlarged system still satisfies Hermite–Lindemann, Baker, Gel'fond–Schneider, six exponentials, Roy's strong six exponentials and four exponentials in transcendence degree one, and it is stable under conjugation. Yet it contains a candidate. So Diaz's conjecture does not follow from these theorems used as properties of $(\mathcal{L}, \widetilde{\mathcal{L}})$.

None of this proves Diaz's conjecture. Theorem C, Corollary 2.2(b), the inversion lemma, all of Theorem D and Theorem E with its corollary are formalised, and so is the answer to Theorem E's question for every $z \in \overline{\mathbb{Q}}(u)$ (Section 9).

1 Setting

Write $\overline{\mathbb{Q}}$ for the algebraic numbers in $\mathbb{C}$, $\mathcal{L} = \{\lambda \in \mathbb{C} : e^\lambda \in \overline{\mathbb{Q}}\}$ for the logarithms of algebraic numbers, and $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$ for the $\overline{\mathbb{Q}}$-vector space spanned by 1 and $\mathcal{L}$. Both are stable under complex conjugation, since $\overline{e^\lambda} = e^{\bar{\lambda}}$. A *candidate* is a $u \neq 0$ with $|u| \in \overline{\mathbb{Q}}$ and $e^u \in \overline{\mathbb{Q}}$; Diaz's conjecture says there is none [Dia04, §5.1]. A candidate is transcendental (Hermite–Lindemann), and

$$u,\ \bar{u} \in \mathcal{L}, \qquad \rho := u\bar{u} = |u|^2 \in \overline{\mathbb{Q}}, \qquad \bar{u} = \rho/u.$$

Definition 1.1. Let $K \subseteq \Omega$ be fields and $V \subseteq \Omega$ a K-vector space. A $p \times q$ *configuration over K in V* is a pair of families $x_1, \dots, x_p$ and $y_1, \dots, y_q$ in Ω, each linearly independent over K, with every product $x_i y_j$ in V. Usually $\Omega = \mathbb{C}$; in Section 3, $\Omega = \overline{\mathbb{Q}}(T)$. Unqualified, "configuration" means a 2×3 configuration over $\overline{\mathbb{Q}}$.

The only transcendence input of Sections 2–6 is the following theorem. It is proved [Roy92], but the project carries it as a hypothesis.

Theorem 1.2 (Roy's strong six exponentials theorem [Roy92, Cor. 2])**.** *There is no configuration in $\widetilde{\mathcal{L}}$.*

Fix $u \in \mathbb{C} \setminus \overline{\mathbb{Q}}$ with $\rho = u\bar{u} \in \overline{\mathbb{Q}}$; then u is transcendental, and $\rho > 0$. Put

$$H_0 = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}u^{-1}, \qquad W_z = H_0 + \overline{\mathbb{Q}}z + \overline{\mathbb{Q}}\bar{z} \quad (z \in \mathbb{C}).$$

We call W_z the *conjugation hull* of z. If u is a candidate and $z \in \widetilde{\mathcal{L}}$, then $W_z \subseteq \widetilde{\mathcal{L}}$, because $\widetilde{\mathcal{L}}$ is a conjugation-stable $\overline{\mathbb{Q}}$-space containing $1, u, \bar{u}$. Hence:

> *If W_z carries a configuration and u is a candidate, then $z \notin \widetilde{\mathcal{L}}$, and $e^{\beta z}$ is transcendental for every $\beta \in \overline{\mathbb{Q}}^{\times}$.*

In this case we say that Roy's theorem *excludes z through its hull*. Theorem B of the project [BP, Thm. 10.6] decides this when $\dim W_z = 4$: for any $z \notin H_0$, $H_0 + \overline{\mathbb{Q}}z$ carries a configuration exactly when $z \in H_0 + \overline{\mathbb{Q}}w$ for some $w \in \{u^2, \bar{u}^2\} \cup \{1/(u-a) : a \in \overline{\mathbb{Q}}^{\times}\}$. The project also treats several families with $\dim W_z = 5$. Its research notes, which are not in the repository, classify the conjugation-stable five-dimensional case with a factor of degree two, with vector bundles on $\mathbb{P}^1$ (see Remark 2.4). Their normal form for a ratio of degree two, which does not involve conjugation, is now formal without bundles, and with it the answer for every $z \in \overline{\mathbb{Q}}(u)$ (Section 9). The explicit conjugation-stable families of the notes, for a general involution, are not formalised.

2 Statements

Polynomials in u

Theorem C (Polynomial hulls)**.** *Let $u \in \mathbb{C} \setminus \overline{\mathbb{Q}}$ with $\rho = u\bar{u} \in \overline{\mathbb{Q}}$, and let $P \in \overline{\mathbb{Q}}[X]$ have degree k. Then $W_{P(u)}$ carries a configuration if and only if $k \in \{2, 3\}$. Explicitly, write $P = p_k X^k + \cdots + p_0$:*

(i) *if $k = 2$, the families $(1, u)$ and $(\bar{u}, 1, u)$ form a configuration in $W_{P(u)}$;*

(ii) *if $k = 3$, put $c = p_2/p_3$ and $Q = u^3 + cu^2$; the families $(1, u^2)$ and $(\bar{u},\ u + c,\ \bar{Q})$ form a configuration in $W_{P(u)}$;*

(iii) *if $k \leq 1$ or $k \geq 4$, there is no configuration in $W_{P(u)}$.*

Corollary 2.1 (Quadratics and cubics are excluded)**.** *Let u be a candidate and $P \in \overline{\mathbb{Q}}[X]$ with $\deg P \in \{2, 3\}$. Then $P(u), P(\bar{u}) \notin \widetilde{\mathcal{L}}$; so $e^{\beta P(u)}$ and $e^{\beta P(\bar{u})}$ are transcendental for every $\beta \in \overline{\mathbb{Q}}^{\times}$.*

Credit. Corollary 2.1 and parts (i)–(ii) are printed. They are one substitution in Diaz's Théorème 7(1) [Dia07], and already in [Dia04, Th. 2], which contains it as the case $x = (v, \bar{v})$. Théorème 7(1) says: if $(v, \bar{v})$ and $(1, w, \bar{w})$ are each free over $\overline{\mathbb{Q}}$, then v, vw and $v\bar{w}$ are not all in $\widetilde{\mathcal{L}}$. Take $v = u$ and $w = u^2 + cu$. Then $v\bar{w} = \rho\bar{u} + \bar{c}\rho \in H_0$, so $vw = Q \notin \widetilde{\mathcal{L}}$; and $w = u + b$ gives the quadratics. The configuration of (ii) is Diaz's $(v, \bar{v}) \otimes (1, w, \bar{w})$, with the first family multiplied by u/ρ and the second by ρ/u. The project has formalised the theorem (`Diaz.Mirror`, `diaz_2007_th7`, part 1), and it also gives the exclusion behind the project's Theorem 10.52 [BP], the case $c = \pm|u|$. The part of Theorem C not found in the sources read is (iii), with its proof (Section 4). The case $P = X^k$ of (iii) is the project's power hull theorem [BP, Thm. 10.14].

Corollary 2.2 (A single polynomial tie of degree at least four)**.** *Let $u \in \mathbb{C} \setminus \overline{\mathbb{Q}}$ with $u\bar{u} \in \overline{\mathbb{Q}}$.*

(a) *For $P \in \overline{\mathbb{Q}}[X]$ of degree at least 4 there is no configuration in $W_{P(u)}$, nor in $W_{P(u)} + \overline{\mathbb{Q}}w_1 + \cdots + \overline{\mathbb{Q}}w_m$ whenever $u, w_1, \ldots, w_m$ are jointly algebraically independent over $\overline{\mathbb{Q}}$.*

(b) *For every polynomial G with real* algebraic *coefficients, the number $z = G(\operatorname{Re} u, \operatorname{Im} u)$ has no configuration in its hull W_z.*

So at a candidate, Roy's theorem cannot exclude a single z such as u^4+u^2, $(\operatorname{Re} u)^2$, $\operatorname{Re} u \cdot \operatorname{Im} u$, $\operatorname{Re}(u^2)$ or $(\operatorname{Im} u)^3$ through its own hull. Several ties together can still be excluded. The project's Proposition 10.30 and Corollary 10.32 [BP] show that u^k and u^l are not both in $\widetilde{\mathcal{L}}$ when $4 \le k < l$ and $l \in \{k+1, k+2, 2k-1, 2k, 2k+1, 3k\}$. For example, $u^4, u^5 \in \widetilde{\mathcal{L}}$ would give the configuration $(1, u) \otimes (u^{-5}, u^{-1}, 1)$. In (b) the coefficients must be algebraic. Every real number is a constant polynomial, and $i/(u-s) + i/(2s)$, with $s = |u|$, is real and excluded by Theorem B.

Polynomials in u and $\bar{u}$

Put $s = |u|$, which is real algebraic, and $v = u/s$, so that $|v| = 1$ and $\bar{v} = v^{-1}$. Every $z \in \overline{\mathbb{Q}}[u, \bar{u}]$ is $L(v)$ for exactly one Laurent polynomial $L \in \overline{\mathbb{Q}}[T, T^{-1}]$, and $\bar{z} = L^\sigma(v)$, where $L^\sigma(T) = \bar{L}(T^{-1})$. Let $H = \operatorname{span}(T^{-1}, 1, T)$, so that $H(v) = H_0$. Let π delete the monomials $T^{-1}, 1, T$, and put

$$M_z = \pi\big(\operatorname{span}(L, L^\sigma)\big), \qquad \text{so} \qquad W_z = (H \oplus M_z)(v), \quad \sigma(M_z) = M_z, \quad \dim W_z = 3 + \dim M_z.$$

Theorem E (Laurent hulls)**.** *In this setting, W_z carries a configuration if and only if* $\dim M_z = 2$ *and M_z belongs to one of the following classes.*

(P) $M_z \cap \operatorname{span}(T^2, T^3) \neq 0$*; equivalently, $W_z = W_{P(u)}$ for some $P \in \overline{\mathbb{Q}}[X]$ of degree 2 or 3 (Theorem C).*

(Q) $M_z \subseteq \operatorname{span}(T^{-3}, T^{-2}, T^2, T^3)$ *and* $\dim\big(M_z \cap \operatorname{span}(T^{-2}, T^2)\big) = 1$*. Equivalently,*

$$M_z = \operatorname{span}\big(T^{-2} + eT^2,\ \beta T^{-3} + \zeta T^2 + eT^3\big), \qquad |e| = |\beta| = 1, \quad \zeta = -\beta e^2 \bar{\zeta}.$$

A configuration is $(1,\, T + \beta T^{-1}) \otimes (1,\, T^{-1} + e\beta T,\, T^{-2} + \zeta T + eT^2)$.

(R) $M_z = \pi\big(\operatorname{span}(g, g^\sigma)\big)$ *with $g = G + \kappa T^{-1} G^\sigma$ for some $G \in T\,\overline{\mathbb{Q}}[T]$ of degree at least 2 and some $|\kappa| = 1$. Then $Tg = \kappa g^\sigma$, and a configuration is $(1, T) \otimes (T^{-1}, 1, g)$. These are the project's real pencil spaces [BP, Prop. 10.42, Thm. 10.43] whose data are Laurent polynomials. In terms of u: W_z contains some $w \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\bar{u}$ with $uw \in \overline{\mathbb{Q}}\bar{w}$, and $W_z \neq W_{u^2}$.*

The three classes are disjoint. When $\dim W_z = 4$ *there is never a configuration.*

Corollary 2.3 (Explicit criteria)**.** *In the setting of Theorem E, suppose* $\dim W_z = 5$.

(a) *If πL has an exponent of absolute value at least 4, then W_z carries a configuration if and only if it is in class (R).*

(b) *Suppose $z \equiv A\bar{u}^3 + B\bar{u}^2 + Cu^2 + Du^3 \pmod{H_0}$. Put $N = |A|^2 - |D|^2$ and $\Delta = A\bar{C} - B\bar{D}$. Then W_z is in class (P) iff $\Delta = 0$; in class (Q) iff $N = 0$ and $(A, D) \neq (0, 0)$; and in class (R) iff $|\Delta| = |u|\,|N| \neq 0$. So W_z carries a configuration if and only if*

$$N \cdot \Delta \cdot \big(|\Delta|^2 - |u|^2 N^2\big) = 0.$$

At a candidate, every z in classes (P), (Q) or (R) lies outside $\widetilde{\mathcal{L}}$. Examples are $\operatorname{Re}(u^3) + cu^2$ for $c \in \overline{\mathbb{Q}}^\times$, which is in class (Q) with $A = D = \frac{1}{2}$ and $B = 0$, $C = c$; and $(u/s)^3 + (\bar{u}/s)^4$, which is in class (R). By contrast, $\operatorname{Re}(u^3)$ alone (the case $c = 0$) is real, and by Corollary 2.2(b) it cannot be excluded.

Remark 2.4 (Class (Q) and the project's research notes)**.** In class (Q) the factor $(1, T+\beta T^{-1})$ is a ratio of degree two. Its involution $T \mapsto \beta/T$ commutes with σ, since $|\beta| = 1$, and exchanges 0 and ∞. The project's research notes classify the conjugation-stable invisible spaces with a factor of degree two into three families, for every such involution; the companion note [CN] and the README give a formal member of each, all for $T \mapsto -T$. Only the normal form behind this classification, which does not involve conjugation, is formal (Section 9); the three families are not. Class (Q) is the cyclic family for $T \mapsto \beta/T$ on its orbit $\{0, \infty\}$: with $h = T + \beta T^{-1}$, $\varphi_0 = \beta^{-1}T^{-1} + eT$ and $\varphi_1 = \zeta T$, one has $T^{-2} + eT^2 \equiv \varphi_0 h$ and $\beta T^{-3} + \zeta T^2 + eT^3 \equiv \varphi_0 h^2 + \varphi_1 h$ modulo H. Its point $\zeta = 0$, $e = \beta^{-2}$, where $M_z = \pi \operatorname{span}(h^2, h^3)$ and the configuration is a progression, is the involution space of [BP, Prop. 10.38, Thm. 10.40] for the pencil $(T^2 + \beta, T)$. The notes only sketch the local cases at 0 and ∞. Theorem E settles that case: such a space has no factor of degree one and a single involution, and all its spaces have Laurent data. A (Q) space is conjugation-stable and invisible to Theorem B. For $z = \operatorname{Re} u \operatorname{Im} u + i(\operatorname{Re} u)^2 \operatorname{Im} u$, exact computation gives $\dim(W_z \cap T^{-2}W_z) = 1$ and $\dim(W_z \cap T^{-1}W_z) = 2$. So that hull has no factor $(1, u^2)$ and no factor of degree one. As exclusions at a candidate, (Q) and (R) are substitutions in Roy's theorem. For (Q) they are also substitutions in [Dia04, Théorème 2], the case $x_1/\bar{x}_1, x_2/\bar{x}_2 \in \overline{\mathbb{Q}}$, and in [Dia07, Théorème 4(1)], both with $x = (1, (u + \beta\bar{u})/s)$, for which $x_2/\bar{x}_2 = \beta$. What was not found in the sources read is the "only if" direction of Theorem E.

A barrier

Theorem D (A generic candidate satisfies (D2)–(D4))**.** *Let F be the subfield of $\mathbb{C}$ generated by $\overline{\mathbb{Q}}$ and $\mathcal{L}$; it is countable and stable under conjugation. Let $\rho > 0$ be real algebraic, and let u be a point of the circle $|u|^2 = \rho$ that is transcendental over F. Such points exist, since the circle is uncountable and the algebraic closure of F is countable. Put*

$$\mathcal{L}_u = \mathcal{L} + \mathbb{Q}u + \mathbb{Q}\bar{u}, \qquad \widetilde{\mathcal{L}}_u = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_u = \widetilde{\mathcal{L}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}.$$

Then:

(D1) *$\mathcal{L}_u \supseteq \mathcal{L}$ is a $\mathbb{Q}$-vector space stable under conjugation. It contains u and $\bar{u}$, with $u \notin \overline{\mathbb{Q}}$ and $u\bar{u} = \rho \in \overline{\mathbb{Q}}$; so u is a candidate for $\mathcal{L}_u$.*

(D2) (Baker) *If $\lambda_1, \dots, \lambda_n \in \mathcal{L}_u$ are linearly independent over $\mathbb{Q}$, then $1, \lambda_1, \dots, \lambda_n$ are linearly independent over $\overline{\mathbb{Q}}$. In particular $\mathcal{L}_u \cap \overline{\mathbb{Q}} = \{0\}$ (Hermite–Lindemann). Also, if $\lambda, \beta\lambda \in \mathcal{L}_u$ with $\lambda \neq 0$ and $\beta \in \overline{\mathbb{Q}}$, then $\beta \in \mathbb{Q}$ (Gel'fond–Schneider).*

(D3) (Roy) *There is no configuration in $\widetilde{\mathcal{L}}_u$.* (Six exponentials) *There is no 2×3 configuration over $\mathbb{Q}$ in $\mathcal{L}_u$.*

(D4) (Four exponentials in transcendence degree one) *Let N be a singular 2×2 matrix whose entries are non-zero, lie in $\mathcal{L}_u$, and generate a field of transcendence degree at most one over $\mathbb{Q}$. Then the rows of N, or its columns, are linearly dependent over $\mathbb{Q}$.*

So the system $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$ is conjugation-stable, satisfies (D2)–(D4), and contains a candidate. Hence *no derivation that uses only (D2)–(D4) and conjugation, as axioms about $(\mathcal{L}, \widetilde{\mathcal{L}})$, can prove Diaz's conjecture.* These theorems are the main transcendence inputs of the project's unconditional results about candidates and of Diaz's papers [Dia04, Dia07]. Theorem D does not cover statements about the values $e^{x_i y_j}$ themselves, such as Waldschmidt's theorem of 1973. Nor does it cover the rank bounds for larger matrices (Remark 7.3). The strong four exponentials conjecture fails in $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$, as it must: the certificate $(1, u) \otimes (1, \bar{u})$ is a 2×2 configuration in $\widetilde{\mathcal{L}}_u$.

3 Tools

Since u is transcendental, $f \mapsto f(u)$ is an injective homomorphism from $\overline{\mathbb{Q}}(T)$ into $\mathbb{C}$. Conjugation corresponds to the semilinear involution

$$f \longmapsto f^\sigma, \qquad f^\sigma(T) = \bar{f}(\rho/T),$$

where $\bar{f}$ has conjugated coefficients. Indeed $\overline{f(u)} = \bar{f}(\bar{u}) = \bar{f}(\rho/u)$. We use the valuations ord_0 and $\mathrm{ord}_\infty = -\deg$ of $\overline{\mathbb{Q}}(T)$. For a Laurent polynomial $f = \sum_j f_j T^j \neq 0$, $\mathrm{ord}_0 f$ is its smallest exponent and $\deg f$ its largest, and its *support* is the interval $[\mathrm{ord}_0 f, \deg f]$.

Lemma 3.1 (Orders of a subspace)**.** *Let ν be ord_0 or ord_∞, and let $V \subseteq \overline{\mathbb{Q}}(T)$ be a $\overline{\mathbb{Q}}$-subspace of finite dimension. Then $\nu(V \setminus \{0\})$ has exactly $\dim V$ elements.*

Proof. The residue field of ν is $\overline{\mathbb{Q}}$. If $f, g \in V$ have the same value n, some $c \in \overline{\mathbb{Q}}^\times$ makes $\nu(f - cg) > n$. So each quotient $V_{\geq n}/V_{>n}$ of the filtration by values has dimension at most one, and it is non-zero exactly when n is a value. This is the valuation argument of [BSZ17, §7, proof of Th. 33] that the project uses for its Lemma 10.17 in $\overline{\mathbb{Q}}[T]$. □

For a configuration $x \otimes y$ write $\mathbf{X} = \mathrm{span}(x_1, x_2)$ and $\mathbf{Y} = \mathrm{span}(y_1, y_2, y_3)$. For $f \in \mathbf{X}$ and $g \in \mathbf{Y}$ the product fg is a combination of the $x_i y_j$, so it lies in V. With Lemma 3.1 this gives:

Lemma 3.2 (Sumsets)**.** *Let $x \otimes y$ be a configuration over $\overline{\mathbb{Q}}$ in a $\overline{\mathbb{Q}}$-space $V \subseteq \overline{\mathbb{Q}}(T)$, with $x_i, y_j \in \overline{\mathbb{Q}}(T)$, and let $\nu = \mathrm{ord}_0$ or ord_∞. Then $A = \nu(\mathbf{X} \setminus 0)$ and $B = \nu(\mathbf{Y} \setminus 0)$ have 2 and 3 elements, and $A + B \subseteq \nu(V \setminus 0)$.*

Lemma 3.3 (Normalisation)**.** *Let $V \subseteq \overline{\mathbb{Q}}[T, T^{-1}]$ and let $x \otimes y$ be a configuration in $V(u) = \{f(u) : f \in V\}$. Then there is a configuration $x' \otimes y'$ with the same six products, such that $x'_i = f_i(u)$ and $y'_j = g_j(u)$ for Laurent polynomials $f_i, g_j \in \overline{\mathbb{Q}}[T, T^{-1}]$.*

Proof. Products of non-zero numbers are non-zero, so $x_1 y_1 \neq 0$. The numbers $x_2/x_1 = x_2 y_1/(x_1 y_1)$ and $x_1 y_j$ lie in $\overline{\mathbb{Q}}(u)$. Replacing x_i by x_i/x_1 and y_j by $x_1 y_j$ changes neither the products nor the independence. So we may assume $x_i = f_i(u)$ and $y_j = g_j(u)$ with $f_i, g_j \in \overline{\mathbb{Q}}(T)$, and $f_i g_j \in V$. Every place of $\overline{\mathbb{Q}}(T)$ other than $0, \infty$ is $T = p$ with $p \in \overline{\mathbb{Q}}^\times$. Let $m_p = \min(\mathrm{ord}_p f_1, \mathrm{ord}_p f_2)$, which is 0 for all but finitely many p, and let $\phi = \prod_p (T - p)^{-m_p}$. Then $\phi f_1, \phi f_2$ have no pole on $\overline{\mathbb{Q}}^\times$. At each p one of them, say ϕf_i, has order 0. Since $(\phi f_i)(\phi^{-1} g_j) = f_i g_j \in V$ has no pole at p, neither has $\phi^{-1} g_j$. Rational functions without poles on $\overline{\mathbb{Q}}^\times$ are Laurent polynomials. Replace f_i, g_j by $\phi f_i, \phi^{-1} g_j$. □

Lemma 3.4 (A rigid five-element set)**.** *Let $k \geq 4$ and $2 \leq m \leq k$, and let $S = \{-k, -1, 0, 1, m\}$. If $A + B \subseteq S$ with $|A| = 2$ and $|B| = 3$, then $m = 2$ and $A + B = \{-1, 0, 1, 2\}$.*

Proof. Write $A = \{a, a + d\}$ with $d \geq 1$. The three numbers $a + b$, $b \in B$, satisfy $s \in S$ and $s + d \in S$, so the difference d occurs at least three times among pairs of elements of S. The ten positive differences of S are

$$1,\ 1,\ 2,\ m-1,\ m,\ m+1 \qquad \text{and} \qquad k-1,\ k,\ k+1,\ k+m,$$

split into three groups. In $G = \{k-1, k, k+1, k+m\}$ the values are distinct and at least 3. In $\{m-1, m, m+1\}$ they are distinct. The rest is $\{1, 1, 2\}$. A value $d \geq 3$ occurs at most once in each of the first two groups and not in the third, so at most twice. The value 2 occurs once in the third group, at most once in the second and never in G. The value 1 occurs twice in the third group, never in G, and in the second group only as $m - 1 = 1$. So a difference occurs three times only if $d = 1$ and $m = 2$. The three pairs are then $(-1, 0), (0, 1), (1, 2)$, so $a + B = \{-1, 0, 1\}$ and $A + B = \{-1, 0, 1, 2\}$. □

The lemma was also checked by machine for every $4 \leq k < 80$ and $2 \leq m \leq k$.

4 Proof of Theorem C

Write $P(u) \equiv p_2u^2 + p_3u^3 + \cdots + p_ku^k \pmod{H_0}$. Conjugation fixes H_0, so $W_{P(u)}$ is spanned by H_0, by $P(u)$ and by its conjugate.

Degree at most one. Then $W_{P(u)} = H_0 = \mathrm{span}(1, T, T^{-1})(u)$ has dimension 3. By Lemma 3.3 a configuration could be taken in $\overline{\mathbb{Q}}[T, T^{-1}]$. Then Lemma 3.2 gives at least $|A + B| \geq |A| + |B| - 1 = 4$ orders at 0 inside a three-dimensional space, which is impossible.

Degree two. Here $u^2 \in W_{P(u)}$, because $P(u) - p_2u^2 \in H_0$. The products of $(1, u)$ and $(\bar{u}, 1, u)$ are $\bar{u}, 1, u, \rho, u, u^2$. Both families are free because u is transcendental and $\bar{u} = \rho/u$.

Degree three. Here $Q = u^3 + cu^2 \equiv P(u)/p_3$ modulo H_0, so Q and $\bar{Q}$ lie in $W_{P(u)}$. The six products of $(1, u^2)$ and $(\bar{u}, u + c, \bar{Q})$ are

$$\bar{u}, \qquad u + c, \qquad \bar{Q}, \qquad u^2\bar{u} = \rho u, \qquad u^2(u + c) = Q, \qquad u^2\bar{Q} = u^2\bar{u}^3 + \bar{c}\,u^2\bar{u}^2 = \rho^2\bar{u} + \bar{c}\rho^2,$$

all in $W_{P(u)}$. The family $(1, u^2)$ is free. So is the family $(\bar{u}, u + c, \bar{Q})$, the value at u of $(\rho T^{-1},\ T + c,\ \rho^3T^{-3} + \bar{c}\rho^2T^{-2})$. These Laurent polynomials have distinct orders at $T = 0$: -1; then 0, or 1 if $c = 0$; then -3. So they are independent, and evaluation at u is injective.

Degree at least four. Let $\mathcal{W} \subseteq \overline{\mathbb{Q}}[T, T^{-1}]$ be spanned by

$$1, \quad T, \quad T^{-1}, \quad L = \sum_{j=2}^{k} p_jT^j, \quad L^\sigma = \sum_{j=2}^{k} \bar{p}_j\rho^jT^{-j}.$$

Then $\mathcal{W}(u) = W_{P(u)}$. Let $m = \min\{j \geq 2 : p_j \neq 0\}$. An element $h + \alpha L + \beta L^\sigma$ with $h \in \mathrm{span}(1, T, T^{-1})$ has order $-k$ at 0 if $\beta \neq 0$. Otherwise its order is $\min(\mathrm{ord}_0 h, m)$, or $\mathrm{ord}_0 h$ when $\alpha = 0$. Hence

$$\mathrm{ord}_0(\mathcal{W} \setminus 0) = \{-k, -1, 0, 1, m\}, \qquad \deg(\mathcal{W} \setminus 0) = \{-m, -1, 0, 1, k\}.$$

The second set follows from the first by the symmetry $f \mapsto f^\sigma$, which exchanges ord_0 and $-\deg$.

Suppose $x \otimes y$ is a configuration in $W_{P(u)}$. By Lemma 3.3 we may take $\mathbf{X}, \mathbf{Y} \subseteq \overline{\mathbb{Q}}[T, T^{-1}]$ with $\mathbf{XY} \subseteq \mathcal{W}$. Let A_0, B_0 be the sets of orders at 0 of $\mathbf{X}$ and $\mathbf{Y}$, and D_A, D_B their sets of degrees. By Lemmas 3.2 and 3.4,

$$A_0 + B_0 = \{-1, 0, 1, 2\}.$$

Applying the same lemmas to $-\deg$, whose values on $\mathcal{W}$ form the same set $\{-k, -1, 0, 1, m\}$, gives

$$D_A + D_B = \{-2, -1, 0, 1\}.$$

Every $f \in \mathbf{X}$ is supported in $[\min A_0, \max D_A]$, and every $g \in \mathbf{Y}$ in $[\min B_0, \max D_B]$. So every product fg is supported in

$$[\min A_0 + \min B_0,\ \max D_A + \max D_B] = [-1, 1].$$

But the product of $f \in \mathbf{X}$ of order $\max A_0$ with $g \in \mathbf{Y}$ of order $\max B_0$ has order $\max(A_0 + B_0) = 2$ at 0. This is a contradiction, and Theorem C is proved. □

Remark 4.1. The proof is two-sided: Lemma 3.4 alone allows $m = 2$. The extra step is that the orders at the two ends of a Laurent polynomial pinch its support. In the monomial case $P = X^k$, the set of orders at 0 already fails for $k \geq 4$, and the proof reduces to the project's power hulls [BP, Thm. 10.14]. Theorem C thus extends the negative half of that result from monomials to all polynomials. The positive half for cubics is Diaz's (Section 2).

5 Proofs of the corollaries of Theorem C

Proof of Corollary 2.1. If $P(u) \in \widetilde{\mathcal{L}}$, then $W_{P(u)} \subseteq \widetilde{\mathcal{L}}$ (Section 1). By Theorem C, $W_{P(u)}$ carries a configuration, which contradicts Theorem 1.2. Next, $P(\bar{u}) = \overline{\bar{P}(u)}$, where $\bar{P}$ has conjugated coefficients and the same degree, and $\widetilde{\mathcal{L}}$ is stable under conjugation; so $P(\bar{u}) \notin \widetilde{\mathcal{L}}$ as well. Finally, if $e^{\beta P(u)}$ were algebraic, then $\beta P(u) \in \mathcal{L} \subseteq \widetilde{\mathcal{L}}$, and so $P(u) \in \widetilde{\mathcal{L}}$. □

Proof of Corollary 2.2. (a) The first claim is Theorem C(iii). For the second, put $F = \overline{\mathbb{Q}}(u)$, which contains $V_0 = W_{P(u)}$, because $\bar{u} = \rho/u$. The w_j are algebraically independent over F. The separation theorem of the project [BP, Thm. 10.16] puts every 2×3 configuration in $V_0 + \sum_j \overline{\mathbb{Q}} w_j$ inside V_0, where there is none.

(b) Since G has real coefficients, z is real. Also $\operatorname{Re} u = (u + \rho u^{-1})/2$ and $\operatorname{Im} u = (u - \rho u^{-1})/(2i)$, so $z = \ell(u)$ for a Laurent polynomial $\ell \in \overline{\mathbb{Q}}[T, T^{-1}]$. If $z \in H_0$, then $W_z = H_0$ carries nothing. Otherwise $W_z = H_0 + \overline{\mathbb{Q}} z$ has dimension 4. If it carried a configuration, Theorem B [BP, Thm. 10.6] would give $z = h + cw$ with $h \in H_0$, $c \in \overline{\mathbb{Q}}^{\times}$, and w equal to u^2, $\bar{u}^2$ or $1/(u - a)$ with $a \in \overline{\mathbb{Q}}^{\times}$. The last is impossible, because $cw = z - h$ would be a Laurent polynomial in u while $1/(T - a)$ has a pole at $a \neq 0$. If $w = u^2$, conjugate $z \equiv cu^2$ modulo H_0; since z is real and H_0 is conjugation-stable, this gives $z \equiv \bar{c}\rho^2 u^{-2}$. Hence $cu^2 - \bar{c}\rho^2 u^{-2} \in H_0$. The monomials $T^{-2}, \dots, T^2$ are independent, so $c = 0$, a contradiction. The case $w = \bar{u}^2$ is symmetric. □

6 Proof of Theorem E

This proof was found in this session three times, independently. The derivations used different methods: valuations, the ratio x_2/x_1, and computation. They agree on the statement, and each was checked adversarially with exact computer algebra (sympy). We give the second in condensed form. Two independent referees have since checked it and found no error; they filled six gaps of exposition (the shape list for every o_1, the list of pencils of Step 0(f), the reduction to the degree bound in Case II, the case $[-3, 1]$ of Step 4, the rank count that replaces the Gröbner basis of Step 6, and the converse of Corollary 2.3(b) for class (R)). The formal proof (Section 9) takes a shorter route: a configuration factors through a coprime pencil of degree at most 2, one count of degrees rules out two of the four pencils, and the other two give (R) and (Q). Write $W = H \oplus M$ with $M = M_z$, and $\mathbf{X}, \mathbf{Y}$ for the spans of a configuration.

Step 0: reductions. (a) By Lemma 3.3, $\mathbf{X}, \mathbf{Y} \subseteq \overline{\mathbb{Q}}[T, T^{-1}]$ with $\mathbf{XY} \subseteq W$. (b) $\dim M \leq 1$ carries nothing. If $M = 0$, argue as for degree at most one in Theorem C. If $M = \overline{\mathbb{Q}} m$, Theorem B puts T^2, T^{-2} or some $1/(T - a)$ in W. The last is not a Laurent polynomial. The first two force $m \propto T^{\pm 2}$, but then $m^{\sigma} \propto T^{\mp 2} \notin \overline{\mathbb{Q}} m$, against $\sigma M = M$. So $\dim M = 2$. (c) Let $O = \operatorname{ord}_0(M \setminus 0) = \{o_1 < o_2\}$, with $o_i \notin \{-1, 0, 1\}$. Since σ exchanges ord_0 and $-\deg$, the degrees of M form $-O$. Supports in H and M are disjoint, so ord_0 and $-\deg$ take the same five values $S = \{o_1, o_2, -1, 0, 1\}$ on $W \setminus 0$. If $o_1 \geq 2$, then $M \subseteq T^2 \overline{\mathbb{Q}}[T]$ and $\sigma M \subseteq T^{-2} \overline{\mathbb{Q}}[T^{-1}]$, so $M = 0$; hence $o_1 \leq -2$. An element of order o_2 has degree in $-O$, so $o_2 \leq -o_1$. (d) *Width.* If $o_2 \leq -2$, every $w \in W \setminus H$ has $\operatorname{ord}_0 w \in O$ and $\deg w \in -O$. (e) *The ratio has degree at most two.* Let $h = x_2/x_1$. We claim $[\overline{\mathbb{Q}}(T) : \overline{\mathbb{Q}}(h)] \leq 2$. Suppose first the products span four dimensions (Case I). Then they form a progression b, bh, bh^2, bh^3 [BSZ17, Lemma 4] (the project's [BP, Lemma 10.2]). That span meets H in dimension at least 2, and the ratio of two independent elements of the intersection has degree at most 2 in T and lies in $\overline{\mathbb{Q}}(h) \setminus \overline{\mathbb{Q}}$. Otherwise the products span all of W (Case II), and [BP, Thm. 10.58] applies. (f) *Pencils.* Write $h = p/q$ with p, q coprime of degree at most 2. At every $a \in \overline{\mathbb{Q}}^{\times}$, p or q is non-zero, so replacing $(\mathbf{X}, \mathbf{Y})$ by $(\operatorname{span}(p, q), (x_1/q)\mathbf{Y})$ keeps $\mathbf{Y}$ inside the Laurent polynomials. After shifting by a power of T and a change of basis, $\operatorname{span}(p, q)$ is one of the following. For each we list the orders

at 0 and the values of $-\deg$:

$$\begin{array}{lll}
T_1 = \mathrm{span}(1, T) & \{0,1\} & \{0,-1\} \\
T_2 = \mathrm{span}(1, T^2) & \{0,2\} & \{0,-2\} \\
T_3 = \mathrm{span}(T^2, T+\beta),\ \beta \neq 0 & \{0,2\} & \{-1,-2\} \\
T_4 = \mathrm{span}(1, T^2+\alpha T),\ \alpha \neq 0 & \{0,1\} & \{0,-2\} \\
T_{5a} = \mathrm{span}(T, T^2+\beta),\ \beta \neq 0 & \{0,1\} & \{-1,-2\} \\
T_{5b} = \mathrm{span}(T^2+\gamma T, 1+\zeta T),\ \gamma\zeta(1-\gamma\zeta) \neq 0 & \{0,1\} & \{-1,-2\}
\end{array}$$

Conjugation maps a configuration in W to one in W, and exchanges T_3 and T_4. A ratio of degree one means pencil T_1.

Step 1: shapes. Lemma 3.2 at 0 and at ∞ requires the gap d of A_0, and that of A_∞, to occur at least three times as a difference in S. This holds at both ends with the same S. Counting differences, with $o_1 \leq -2$ and $o_2 \leq -o_1$, leaves the following shapes O. The count was checked by machine for $|o_1| \leq 40$.

- $\{-2,2\}$, $\{-3,2\}$, $\{-3,3\}$, and $\{-3,-2\}$.
- $\{-k,2\}$ with $k \geq 4$; here $d=1$ and $a + B_0 = \{-1,0,1\}$.
- $\{o_1,-2\}$ with $o_1 \leq -5$; here $d=1$ and $a + B_0 = \{-2,-1,0\}$.
- $\{c, c+1\}$ with $c \leq -4$; here $d=1$ and $a + B_0 = \{c,-1,0\}$.
- $\{-4,-2\}$; here $d=1$ with $\{-2,-1,0\}$, or $d=2$ with $\{-4,-2,-1\}$.
- $\{-5,-3\}$; here $d=2$.

Step 2: the two-sided pinch. For $\{-k,2\}$ the argument of Theorem C applies. Products are supported in $[-1,1]$, yet one has order 2. Now take $\{o_1,-2\}$, or $\{-4,-2\}$ with $d=1$ at both ends. Products are supported in $[-2,2]$, so they lie in $H \oplus (M \cap \mathrm{span}(T^{-2}, T^2))$, and they span at least four dimensions. That intersection is then a conjugation-stable line, $\overline{\mathbb{Q}}(T^{-2} + eT^2)$ with $e \neq 0$. The products therefore form a configuration in a four-dimensional extension of H that contains neither T^2 nor T^{-2}, against Theorem B.

Step 3: widths. For $\{-5,-3\}$, gap 2 at both ends forces T_2. Then $\mathbf{Y} \subseteq W$ and $T^2\mathbf{Y} \subseteq W$, so every $y \in \mathbf{Y}$ has degree at most $5-2=3$. Its M-part therefore lies in the line of elements of M of degree at most 3. Moreover $\mathbf{Y} \cap H$ has orders in $B_0 \cap \{-1,0,1\} = \{-1\}$, so it has dimension at most one, and $\dim \mathbf{Y} \leq 2$. For $\{-4,-2\}$ with $d=2$ at 0 (the case at ∞ is the conjugate), the pencil is T_2 or T_3. For T_2 the same count applies. For T_3, $B_0 = \{-4,-2,-1\}$ and the end at ∞ forces $\deg y \leq 0$ on $\mathbf{Y}$. Then the element y of order -4 gives $(T+\beta)y \in W$ of order -4 and degree at most 1, against the width property.

Step 4: a ratio of degree one gives (R). Let $V_1 = W \cap T^{-1}W \supseteq \mathbf{Y}$. Then $\sigma V_1 = W \cap TW = TV_1$. If $\dim V_1 \geq 4$, its orders form an interval B with $B + \{0,1\} \subseteq S$, so $S = [-2,2]$ and $O = \{-2,2\}$, which is (P). The other possibility, $[-3,1]$, puts T^2 in M, which is absurd. So $V_1 = \mathrm{span}(T^{-1}, 1, g)$ and $W = \mathrm{span}(T^{-1}, 1, T, g, Tg)$; otherwise Case I contains H, and Theorem B gives (P) again. From $\sigma V_1 = TV_1$ we get $g^\sigma = \alpha + \beta T + \gamma Tg$. Applying σ once more gives $|\gamma| = 1$ and $\alpha = -\gamma\bar{\beta}$. Then $g' = g - \bar{\beta}T^{-1}$ satisfies $g'^\sigma = \gamma T g'$. Comparing coefficients gives $g' \equiv G + \bar{\gamma}T^{-1}G^\sigma$ modulo $\mathrm{span}(1, T^{-1})$, with $G \in T\overline{\mathbb{Q}}[T]$. If $\deg G = 1$, then $T^2 \in M$, which is (P). Otherwise we are in (R), with $\kappa = \bar{\gamma}$.

Step 5: the shape $\{c, c+1\}$ with $c \leq -4$ is (R). Gap 1 at both ends forces T_1 or T_5. Take T_5. Then $B_0 = \{c,-1,0\}$, and the degrees on $\mathbf{Y}$ are $\{-c-2,-1,-2\}$. The subspace of $\mathbf{Y}$ of degree at most -1 is two-dimensional, so it contains some y_1 of order c. With $p_0 \in \mathbf{X}$, $p_0(0) \neq 0$, the product $p_0 y_1 \in W$ has order c and degree at most 1, against the width property. So the pencil is T_1, and Step 4 applies.

Step 6: the shape $\{-3,-2\}$. Here $M \subseteq \mathrm{span}(T^{\pm 2}, T^{\pm 3})$ and $M \cap \mathrm{span}(T^2, T^3) = 0$.

- T_2, T_4, with $h = T^2 + \alpha T$. Any $w = h_0 + m$ with $m \neq 0$ in M has $\deg hw \geq 4$, which is outside W. So $\mathbf{Y} \subseteq H$, and $h \cdot \mathbf{Y} \subseteq W$ then forces $\mathbf{Y} \subseteq \overline{\mathbb{Q}}T^{-1}$. The case T_3 is the conjugate.

- T_{5b}. Then $\mathbf{Y}$ is supported in $[-3,1]$. Write n_2 for the element of M of degree 2 with leading coefficient 1, and n_{-2} for the element of order -2 with lowest coefficient 1. Then $(1+\zeta T)y \in H + \overline{\mathbb{Q}}n_2$ and $(T^2+\gamma T)y \in H + \overline{\mathbb{Q}}n_{-2}$. This gives six linear equations in seven unknowns. We checked by a Gröbner basis that the coefficient matrix has rank at least 5 for every admissible choice of parameters. So $\dim \mathbf{Y} \leq 2$.
- T_{5a}, with $h = T + \beta T^{-1}$. For hw to stay in W, the M-part of each $w \in \mathbf{Y}$ must lie in $M_0 = M \cap \mathrm{span}(T^{-2}, T^2)$. If $M_0 = 0$, then $\dim \mathbf{Y} \leq 1$. Otherwise $M_0 = \overline{\mathbb{Q}}(T^{-2} + eT^2)$ with $|e| = 1$. Modulo H, $h(xT^{-1} + y + zT + t(T^{-2} + eT^2)) \equiv \beta x T^{-2} + zT^2 + t(\beta T^{-3} + eT^3)$. So $\dim \mathbf{Y} \geq 3$ if and only if $M \subseteq \mathrm{span}(T^{-2}, T^2, \beta T^{-3} + eT^3)$. Conjugation-stability then gives $|\beta| = 1$ and $\zeta = -\beta e^2 \bar{\zeta}$: this is (Q).
- T_1 gives (R), by Step 4.

Conversely, let $M \subseteq \mathrm{span}(T^{\pm 2}, T^{\pm 3})$ be conjugation-stable with $\dim(M \cap \mathrm{span}(T^{\pm 2})) = 1$. Then the $T^{\pm 3}$-coefficients a, d of a second generator satisfy $|a| = |d|$, and M normalises to the form in (Q).

Step 7: the shapes $\{-2,2\}$, $\{-3,2\}$, $\{-3,3\}$ are (P). An element of M of order at least 2 has degree at most $-o_1 \leq 3$. So it lies in $\mathrm{span}(T^2, T^3)$, and Theorem C gives the configurations.

Configurations, and disjointness. The configurations listed in (Q) and (R) were checked symbolically for general parameters: all six products lie in W, and both families are free. In (R), the identity $Tg = \kappa g^\sigma$ is a direct computation. On disjointness: (Q) has $M \cap \mathrm{span}(T^2, T^3) = 0$, so it is not (P). Its intersection with $\mathrm{span}(T^{\pm 2})$ is non-zero, while in (R) that intersection is 0 (computed from G), so it is not (R). And (R) with $\deg G \geq 2$ has shape $\{-\deg G - 1, -\deg G\}$, which is not a (P) shape. □

Proof of Corollary 2.3. (a) An exponent of absolute value at least 4 in πL gives $o_1 \leq -4$, because M is conjugation-stable. Of the shapes in Step 1, only $\{c, c+1\}$ with $c \leq -4$ survives Steps 2 and 3, and it is (R). (b) In terms of T, let the coefficients of πL at T^{-3}, T^{-2}, T^2, T^3 be $a = As^3$, $b = Bs^2$, $c = Cs^2$ and $d = Ds^3$. Those of πL^σ are $\bar{d}, \bar{c}, \bar{b}, \bar{a}$. A combination lies in $\mathrm{span}(T^2, T^3)$ iff it kills T^{-3} and T^{-2}, which needs $\det \begin{pmatrix} a & b \\ \bar{d} & \bar{c} \end{pmatrix} = 0$, that is $\Delta = 0$. A combination lies in $\mathrm{span}(T^{\pm 2})$ iff it kills $T^{\pm 3}$, which needs $|a|^2 - |d|^2 = 0$, that is $N = 0$. For (R) the condition is $|\Delta| = s|N| \neq 0$. This was derived by one of the three derivations and re-derived by its referee; it checks against $g = G + \kappa T^{-1} G^\sigma$ with $\deg G = 2$, where $N = |g_2|^2/s^6$ and $|\Delta| = |g_2|^2/s^5$. □

7 Proof of Theorem D

Lemma 7.1 (Adjoining a generic point of a circle)**.** *Let $F \subseteq \mathbb{C}$ be a field containing $\overline{\mathbb{Q}}$, let $V_0 \subseteq F$ be a $\overline{\mathbb{Q}}$-subspace, and let u be transcendental over F. Then every configuration in $V_0 + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}u^{-1}$ has all six products in V_0.*

Proof. As in Lemma 3.3, we may assume $x_i, y_j \in F(u)$; identify $F(u)$ with $F(T)$. Then $Tx_iy_j = P_{ij}(T) = m_{ij} + c_{ij}T + l_{ij}T^2$ with $m_{ij}, l_{ij} \in \overline{\mathbb{Q}}$ and $c_{ij} \in V_0$. The matrix (P_{ij}) has rank one over $F(T)$. Since $F[T]$ is a principal ideal domain, $P_{ij} = a_i b_j$ with $a \in F[T]^2$ *primitive*, meaning $\gcd(a_1, a_2) = 1$, and $b \in F[T]^3$, all entries non-zero. Take a proportional to a non-zero column and divide by the gcd; every column is then a times an element of $F(T)$, and that element is a polynomial because $\gcd(a_1, a_2) = 1$. Then $x_i = \kappa a_i$ and $y_j = \kappa' b_j$ for some $\kappa, \kappa' \in F(T)^\times$ independent of i and j. So the a_i are linearly independent over $\overline{\mathbb{Q}}$, and so are the b_j. Write $a = \sum_r \alpha_r T^r$ and $b = \sum_r \beta_r T^r$ with $\alpha_r \in F^2$ and $\beta_r \in F^3$, and let δ_a, δ_b be the largest degrees of the entries. Then $\delta_a + \delta_b \leq 2$. The constant coefficients give $\alpha_0 \beta_0^{\mathsf{T}} = (m_{ij})$, with entries in $\overline{\mathbb{Q}}$. If $\delta_a + \delta_b = 2$, the coefficients of T^2 give $\alpha_{\delta_a} \beta_{\delta_b}^{\mathsf{T}} = (l_{ij})$, also with entries in $\overline{\mathbb{Q}}$.

We use one observation repeatedly. If $\gamma\delta^{\mathsf{T}}$ is non-zero with entries in $\overline{\mathbb{Q}}$, then γ and δ are each an element of F times a vector with algebraic entries. A vector b of that form has $\overline{\mathbb{Q}}$-dependent entries, and a vector a of that form has $a_2/a_1 \in \overline{\mathbb{Q}}$.

Case $\delta_a = 0$. Then $a \in F^2$, with both entries non-zero. If $\beta_0 \neq 0$, then $a\beta_0^{\mathsf{T}} \neq 0$ has algebraic entries, so $a_2/a_1 \in \overline{\mathbb{Q}}$, which is excluded. Hence $\beta_0 = 0$, and likewise $\beta_2 = 0$. So $b_j = \beta_{j1}T$, and $x_i y_j = a_i\beta_{j1} = c_{ij} \in V_0$.

Case $\delta_a \geq 1$. If $\alpha_0 = 0$, then T divides a, against primitivity. So $\alpha_0 \neq 0$. If $\delta_b = 0$, then $b = \beta_0$ and $\alpha_0\beta_0^{\mathsf{T}} \neq 0$, so b has dependent entries. Otherwise $\delta_a = \delta_b = 1$, and both $\alpha_0\beta_0^{\mathsf{T}}$ (when $\beta_0 \neq 0$) and $\alpha_1\beta_1^{\mathsf{T}} \neq 0$ have algebraic entries. Then every b_j lies in $\overline{\mathbb{Q}}f_0 + \overline{\mathbb{Q}}f_1 T$ for some $f_0, f_1 \in F$, a space of dimension at most two. So the three b_j are dependent. Every case with $\delta_a \geq 1$ is contradictory. □

Take $F = \overline{\mathbb{Q}}(w_1, \dots, w_m)$ and $V_0 = \overline{\mathbb{Q}} + \sum_j \overline{\mathbb{Q}}w_j$, with $u, w_1, \dots, w_m$ algebraically independent. Together with the separation theorem [BP, Thm. 10.16], which leaves no configuration in this V_0, the lemma then gives Theorem A of the project [BP, Thm. 10.1]. The method is that of the project's proof of Theorem A: factor a rank-one polynomial matrix and compare degrees. The separation theorem allows any $V_0 \subseteq F$, but it needs the adjoined numbers to be algebraically independent over F, and u, u^{-1} are not. The lemma does both at once.

Lemma 7.2 (Subsets of transcendence degree one)**.** *In the setting of Theorem D, let* $V \subseteq \mathcal{L}_u$ *be a finite-dimensional* $\mathbb{Q}$*-subspace with* $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(V) \leq 1$*. Then* $V \subseteq \mathcal{L}$*, or* $V \subseteq \mathbb{Q}u + \mathbb{Q}\bar{u}$*, or* $\dim_{\mathbb{Q}} V \leq 1$*.*

Proof. The sum $\mathcal{L}_u = \mathcal{L} \oplus \mathbb{Q}u \oplus \mathbb{Q}\bar{u}$ is direct, since $\lambda + pu + q\rho u^{-1} = 0$ with $\lambda \in F$ forces $p = q = 0$, u being transcendental over F. Let $\pi\colon \mathcal{L}_u \to \mathbb{Q}u \oplus \mathbb{Q}\bar{u}$ be the projection. Two facts are used throughout. A number $\lambda + pu + q\bar{u}$ with $(p, q) \neq (0, 0)$ is transcendental over F. And an element of F that is algebraic over $\mathbb{Q}(e)$, for some e transcendental over F, is algebraic, hence 0 if it lies in $\mathcal{L}$.

If $\dim \pi(V) = 2$, pick $e_1 = \lambda_1 + u$ and $e_2 = \lambda_2 + \bar{u}$ in V. They are algebraically dependent, say $\Phi(e_1, e_2) = 0$ with $\Phi \in \mathbb{Q}[X, Y] \setminus \{0\}$. As u is transcendental over F, Φ vanishes on the hyperbola $(X - \lambda_1)(Y - \lambda_2) = \rho$: the element $\Phi(\lambda_1 + T, \lambda_2 + \rho/T)$ of $F(T)$ vanishes at $T = u$, hence is zero. So the irreducible polynomial $\Psi = (X - \lambda_1)(Y - \lambda_2) - \rho$ divides Φ in $\mathbb{C}[X, Y]$. Irreducible factors over $\overline{\mathbb{Q}}$ stay irreducible over $\mathbb{C}$, so Ψ is a scalar multiple of a polynomial over $\overline{\mathbb{Q}}$. Its coefficient of XY is 1, so Ψ itself has algebraic coefficients. Hence $\lambda_1, \lambda_2 \in \mathcal{L} \cap \overline{\mathbb{Q}} = \{0\}$. Then any $e = \lambda + pu + q\bar{u} \in V$ gives $\lambda = e - pe_1 - qe_2 \in V$. This λ is algebraic over $\mathbb{Q}(V)$, so over $\mathbb{Q}(u)$, and therefore $\lambda = 0$. Thus $V \subseteq \mathbb{Q}u + \mathbb{Q}\bar{u}$.

If $\dim \pi(V) = 1$, pick $e_0 \in V$ with $\pi(e_0) \neq 0$; it is transcendental over F. Every $\lambda \in V \cap \mathcal{L}$ is algebraic over $\mathbb{Q}(e_0)$, so $\lambda = 0$. Hence $V \cap \mathcal{L} = 0$, and $\dim V = \dim \pi(V) = 1$.

If $\pi(V) = 0$, then $V \subseteq \mathcal{L}$. □

Proof of Theorem D. (D1) Conjugation maps $\mathcal{L} + \mathbb{Q}u + \mathbb{Q}\bar{u}$ onto $\mathcal{L} + \mathbb{Q}\bar{u} + \mathbb{Q}u$.

(D2) Let $\lambda_1, \dots, \lambda_n \in \mathcal{L}_u$ be independent over $\mathbb{Q}$, and suppose $\beta_0 + \sum_i \beta_i\lambda_i = 0$ with $\beta_i \in \overline{\mathbb{Q}}$. Changing the $\mathbb{Q}$-basis of $\operatorname{span}_{\mathbb{Q}}(\lambda_i)$, we may assume $\lambda_{r+1}, \dots, \lambda_n$ span the kernel of π, which is $\operatorname{span} \cap \mathcal{L}$, and that $\pi(\lambda_1), \dots, \pi(\lambda_r)$ are independent, with $r \leq 2$. Since $1, u, u^{-1}$ are independent over F, the coefficients of u and $\bar{u}$ in the relation vanish: $\sum_{i \leq r} \beta_i \pi(\lambda_i) = 0$. Vectors of $\mathbb{Q}^2$ independent over $\mathbb{Q}$ remain so over $\overline{\mathbb{Q}}$, so $\beta_i = 0$ for $i \leq r$. What remains is $\beta_0 + \sum_{i>r} \beta_i\lambda_i = 0$, with $\mathbb{Q}$-independent $\lambda_i \in \mathcal{L}$, and Baker's theorem [Bak66] gives $\beta = 0$. Hermite–Lindemann is the case $n = 1$. For Gel'fond–Schneider: if $\beta \notin \mathbb{Q}$, then λ and $\beta\lambda$ are $\mathbb{Q}$-independent, yet $\beta \cdot \lambda - 1 \cdot \beta\lambda = 0$.

(D3) Take $V_0 = \widetilde{\mathcal{L}} \subseteq F$ in Lemma 7.1, with $\bar{u} = \rho u^{-1}$. A configuration in $\widetilde{\mathcal{L}}_u$ would be one in $\widetilde{\mathcal{L}}$, against Theorem 1.2. Now let x and y be $\mathbb{Q}$-independent with all products in $\mathcal{L}_u$. If $x_2 = \beta x_1$ with $\beta \in \overline{\mathbb{Q}}$, then $\beta \notin \mathbb{Q}$ and $x_1 y_1, \beta x_1 y_1 \in \mathcal{L}_u$, against (D2). If $\sum_j \beta_j y_j = 0$ non-trivially, then

the $x_1 y_j \in \mathcal{L}_u$ are $\overline{\mathbb{Q}}$-dependent. By (D2) they are $\mathbb{Q}$-dependent, so y is $\mathbb{Q}$-dependent. Hence x and y are $\overline{\mathbb{Q}}$-independent, and $x \otimes y$ is a configuration in $\widetilde{\mathcal{L}}_u$, against the first part.

(D4) Let V be the $\mathbb{Q}$-span of the entries of N. If $V \subseteq \mathcal{L}$, this is the four exponentials theorem in transcendence degree one, due to Waldschmidt and Brownawell [BP, Thm. 6.31]. If $\dim V = 1$, the entries are $r_{ij}e$ with $r_{ij} \in \mathbb{Q}$, so $\det(r_{ij}) = 0$ and the rows are $\mathbb{Q}$-dependent. If $V \subseteq \mathbb{Q}u + \mathbb{Q}\bar{u}$, the entries are rational linear forms in $(u, \bar{u})$. The determinant is a rational quadratic form $aX^2 + bXY + cY^2$ vanishing at $(u, \rho/u)$. Multiplying by u^2 gives $au^4 + b\rho u^2 + c\rho^2 = 0$, so $a = b = c = 0$. A 2×2 matrix of linear forms whose determinant vanishes identically has dependent rows or columns [BP, Lemma 12.1]. By Lemma 7.2 these cases are exhaustive. □

Remark 7.3 (Scope of Theorem D)**.** The system $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$ has no exponential function. It models statements about membership in $\mathcal{L}$ and $\widetilde{\mathcal{L}}$, not statements about the values $e^{x_i y_j}$. One such statement is Waldschmidt's theorem of 1973 [W73], which the project uses in its "mixed rigidity" [BP, Thm. 7.11]. In that form, x_1, x_2 and y_1, y_2 are $\mathbb{Q}$-linearly independent, and $e^{x_1 y_2}, e^{x_2 y_2}$ are algebraic. Then two of the eight numbers $x_i, y_j, e^{x_i y_j}$ are algebraically independent. A proof of Diaz's conjecture through such a statement is not ruled out here.

Nor is it known whether Roy's rank bound for larger matrices holds in $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$. By [Roy92, Cor. 1] (p. 38; also quoted in [Mak23]), a matrix with entries in $\widetilde{\mathcal{L}}$ has rank at least $m\theta/(1+\theta)$, where θ is computed from its zero blocks after changes of basis. This is a lower bound, not an exact rank. If the bound fails in the model, it can see a generic candidate. We probed it at the first open case, $u^4 \in \widetilde{\mathcal{L}}$ (Section 8).

8 What this means for Diaz's conjecture

The reach of Roy's theorem through one hull. Theorems C and E answer, for every z that is a polynomial in u and $\bar{u}$, whether Roy's strong six exponentials theorem can refute "$z \in \widetilde{\mathcal{L}}$" at a candidate using only z, its conjugate and the certificate H_0. Every exclusion they list is a routine substitution, into Diaz's theorems or into Roy's. What was not found in the sources read is the converse. Apart from classes (P), (Q) and (R) nothing can be excluded this way. For a polynomial in u alone the limit is degree three, a precise form, for one polynomial in u, of Diaz's remark that within the strong six exponentials theorem one cannot hope to go far [Dia07, p. 390]. This concerns *one* hypothetical logarithm at a time. Two ties together can reach further (for example u^4 and u^5; Corollary 2.2). So can additional numbers that are algebraically dependent on u.

Why every known route is conditional. The project derives Diaz's conjecture from several principles: the strong four exponentials conjecture [Wal00, Conj. 11.17], Waldschmidt's strong five exponentials conjecture, and the structural rank conjecture. Each lets algebraic constants enter the products, and each fails in $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$, because the certificate $(1, u) \otimes (1, \bar{u})$ violates it. The theorems in (D2)–(D4) hold there (Theorem D). So a proof must use some property of $(\mathcal{L}, \widetilde{\mathcal{L}})$, or of the exponential function, that fails in $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$. Three kinds are not excluded:

- an inhomogeneous principle of the kind above;
- a statement about the values $e^{x_i y_j}$, such as Waldschmidt's theorem of 1973, or Roy's rank bound for larger matrices if it fails in the model (Remark 7.3);
- the way a genuine candidate, which lies in $\mathcal{L}$ and so is not generic over F, is tied to the other logarithms.

Under Kirby's weak Schanuel conjecture, candidates satisfy $\operatorname{Im} u \in \pi\mathbb{Q}$ [BP, Prop. 13.20]. So u is algebraic over $\mathbb{Q}(\pi) \subseteq F$: such candidates are never generic, and Theorem D says nothing about them. On that branch the four exponentials theorem in transcendence degree one does

give unconditional results; those are the project's Chapter 13. The tie to $i\pi$ lies inside H_0, so it gives Roy's theorem nothing further through hulls.

Higher rank: a reduction and a barrier. Rank-one configurations cannot exclude u^4 (the power hulls). We asked whether Roy's rank bound for larger matrices can. Write $V_\Sigma = \operatorname{span}(T^s : s \in \Sigma)$ for a finite set of integers Σ, and $V = V_{\{-4,-1,0,1,4\}}$, which is the hull of u^4 at $T = u$. A matrix is *graded* if each entry is a constant times $T^{a_i+b_j}$.

The reduction. If a matrix over V_Σ breaks Roy's bound, so does a graded stable matrix over V_Σ of no larger size and rank. Degenerating the matrix itself can create zero blocks, so the limit is taken instead in the projective moduli space of semistable Kronecker modules [Kin94, MFK]: a fixed point of the torus that scales T, on the image of the rank locus, is represented by a direct sum of stable modules, each of them graded. Non-semistable matrices are first split along a block of minimal ratio, an induction modelled on Roy's proof of [Roy92, Th. 5]. This supplies the argument that an earlier version of this note claimed and did not write out (corrected on 23 October 2026).

The barrier. If every non-zero difference of Σ has at most two representations, no matrix of rank at most two over V_Σ breaks the bound; otherwise a 2×3 matrix of rank one does [Roy92, Cor. 2]. A count on incidences in $\mathbb{P}^2$ then shows that *no matrix of rank at most three over V breaks Roy's bound*, for any size, graded or not. So Roy's bound at rank at most three cannot exclude u^4 through its hull. Rank four is open. The graded case has two further checks: a SAT search over the whole proved range $d, l \le 15$, under two solvers, and a separate enumeration up to $d = 8$. Both find no such matrix, and both find the known failures in $\operatorname{span}(T^{-2}, \dots, T^2)$.

Other spaces. The same holds for the hull of $u^4 + cu^3$, $c \neq 0$, modelled by $\operatorname{span}(1, T, T^{-1}, T^4 + cT^3, T^{-4} + c'T^{-3})$ for any c', although its initial space $V_{\{-4,-1,0,1,3\}}$ does carry a failure at rank three: an explicit 5×8 matrix, the only graded stable one with $d \le l$, up to changes of basis and shifts of weights. It does not lift: in a suitable basis, a 4×4 minor of a lift is $2w^2D$ with $w = T^3 + T^4/c$ and $D \equiv -4T^2$ modulo w. Allowing free entries, one indeterminate for each further logarithm, gives no failure at rank at most three either, for V and for the certificate space. That concerns the rank over $\overline{\mathbb{Q}}(T, x)$, not actual logarithms, so it does not answer the question of Remark 7.3. For the certificate space alone, the bound gives nothing at any rank. This is Roy's lemma on spaces of dimension at most three [Roy95, Th. 3.4] (also [Wal00, Exercise 12.10]): rank $\ge (d + l)/4 \ge dl/(d + l)$.

These results use Roy's bound in the quoted form, are not formal, and are not part of the theorems above. The rank-three and hull statements are computer-assisted. Full proofs and the scripts are in the companion note `tex/diaz-roy-rank-three.tex` and `scripts/roy_rank/`. Each part was read by one independent referee agent, and the whole note by a further one who saw only that note; they found no error in a theorem. The statements were not found in the sources read, which did not include Roy's papers of 1989 and 1990 or Atkinson and Lloyd's of 1980, and the reduction assembles standard parts. The good-quotient property of [MFK] was checked only through later papers.

Next steps.

(1) Done: Theorem E and Corollary 2.3 are formal (Section 9).

(2) Done: Theorem E's question is answered for all of $\overline{\mathbb{Q}}(u)$, as the project's Theorem F (formal, 23 October 2026; Section 9). The route is not the one sketched here, one sumset condition per place, but a normal form of the configurations whose ratio has degree two. For Laurent z, Theorem F gives back the classes of Theorem E; this is now proved (Section 9), through Theorem E itself, so Corollary 2.3 still does not follow from Theorem F.

(3) Roy's rank bound for u^4: rank four, and whether the bound holds in the generic model $(\mathcal{L}_u, \widetilde{\mathcal{L}}_u)$, where only T can be scaled and the reduction above does not apply to the

logarithms. The reduction and rank at most three are done (Higher rank, above; not formal).

9 Status and sources

- *Proofs.* Theorems C and D and the corollaries of C are proved here, by hand. The proof of Theorem E is condensed. It was found three times independently in this session, each derivation checked adversarially with exact computer algebra; we re-ran the key symbolic step (T_{5b}) ourselves. Five independent referees checked Theorems C and D. They found no mathematical error, but several overclaims and two misattributions, all corrected in this version.
- *Formalised* (16 October 2026, through Prove2Me, mirrored as `Diaz.Mirror`): Theorem C, as `polynomial_hull_two_by_three_configuration_iff` with part (iii) as `polynomial_hull_no_two_by_three_configuration`; Corollary 2.2(b), as `real_polynomial_hull_no_two_by_three_configuration`; Lemma 7.1, for any subfields $K \subseteq F$, as `rank_one_config_inverse_adjoin`; and (D3) of Theorem D, under Roy's theorem as a hypothesis, as `logAlgTilde_adjoin_circle_point_no_two_by_three_configuration`. The helpers are Lemma 3.4 (`five_point_sumset_rigid`), the degree half of Lemma 3.1 (`polynomial_submodule_degrees_card`), the order sets of Section 4 (`shifted_polynomial_hull_orders`), the normalisation (`config_shifted_polynomial_spans`) and the rank-one factorisation (`rank_one_polynomial_matrix_factor`). Theorem E is formal as `laurent_conj_hull_two_by_three_configuration_iff`, Corollary 2.3(a) as `laurent_conj_hull_large_exponent_configuration_iff_real_pencil_class` and (b) as `cubic_window_conj_hull_two_by_three_configuration_iff_criterion`, and the exclusion at a candidate, under Roy's theorem, as `candidate_laurent_conj_hull_class_not_mem_logAlgTilde` (18 October 2026, 25 results in all). The rest of Theorem D is formal as `logAlg_adjoin_circle_point_theorem_D`, with the reading of Section 8 as `theorem_D_barrier` and the same properties for $\mathcal{L}$ as `span_logAlg_theorem_D_properties` (18 October 2026, 16 results in all). Its six exponentials clause needs neither Roy's theorem nor Gel'fond and Schneider's: Baker's theorem for $\mathcal{L}_u$ covers the dependent cases, and otherwise Lemma 7.1 puts the products in $\mathcal{L}$. Theorem E's question for every $z \in \overline{\mathbb{Q}}(u)$ is formal as the project's Theorem F, `rational_conj_hull_two_by_three_configuration_iff`: W_z carries a configuration exactly when $\dim W_z = 4$ and $z \in \operatorname{span}(1, u, \bar{u}, 1/(u-a))$ with $a\bar{a} = \rho$, or $\dim W_z = 5$ and W_z has one of three shapes (degree one, two-orbit, cyclic). In dimension five conjugation plays no part (`rational_space_two_by_three_configuration_iff`), and the key step is a normal form of the configurations whose ratio has degree two, the statement of Theorem 2 of the research notes proved without vector bundles (Theorem NF, `degree_two_configuration_normal_form`); the degree-one shape is described by `degree_one_multiplier_space_visible_or_degree_one_space`, and `conj_stable_degree_one_multiplier_space_visible_or_real_pencil_space` refines it under conjugation (23 October 2026, 22 results in all). So Theorem E's statement that no four-dimensional hull carries a configuration is special to Laurent z. For Laurent z with $\dim W_z = 5$, the three shapes are exactly the classes (P), (Q) and (R): `laurent_conj_hull_configuration_shape_iff_class`, a corollary of Theorems E and F, with the explicit maps `laurent_polynomial_class_conj_hull_two_orbit_form` ((P) is two-orbit), `laurent_involution_class_space_cyclic_form` ((Q) is cyclic) and `laurent_real_pencil_class_space_u_multiplier_triple` ((R) has the degree-one shape). As the proof goes through Theorem E, Corollary 2.3 still does not follow from Theorem F. For a configuration in a five-dimensional $W \supseteq \operatorname{span}(1, u, u^{-1})$ whose ratio $h = x_2/x_1$ has degree two in u, the products $x_1 y_j$ span exactly $\{f \in W : hf \in W\}$ (`degree_two_configuration_multiplier_space_eq`); and at a candidate, under Roy's theorem, no

$z \in \overline{\mathbb{Q}}(u)$ whose hull meets Theorem F's criterion lies in $\widetilde{\mathcal{L}}$ (`candidate_rational_conj_hull_criterion_not_mem_logAlgTilde`; 23 October 2026, 6 results in all). The explicit conjugation-stable families of the research notes, for a general involution, are not formal.

- *Credit.* The exclusions of quadratics and cubics are Diaz's [Dia07, Théorème 7(1)], formalised in the project. The real pencil class (R) is the project's [BP, Prop. 10.42, Thm. 10.43], and for Laurent data [BP, Lemma 10.59, Thm. 10.60, Prop. 10.61]. The configuration of the project's Theorem 10.52 is also Diaz's [Dia07, Théorème 7(1)]. Its source line in the blueprint, the chapter of results not found in the sources, the README and version 1.29 of the companion note now say so; only its invisibility remains "not found". Not found in the sources read (the project's notes and a web search) are the "only if" halves of Theorems C and E, class (Q) as a family of hulls (its configurations are [Dia04, Théorème 2]), Corollary 2.3, Corollary 2.2(b), and Theorem D; and, outside the research notes, Theorem NF and the "only if" half of Theorem F. Theorem F's four-dimensional configuration is [Dia07, Théorème 7(1)] at $v = 1/(u-a)$, with the exclusion in [Dia07, Corollaire 5(4)]; when the six products span only four dimensions, a configuration is a progression by [BSZ17, Lemma 4]. The six results that close Theorem F's caveats (23 October 2026, above) are routine and were not found in the sources read.
- *Literature.* During this session the network blocked arXiv and most author pages. Web search reached abstracts and quotations only. Roy's 1992 paper was at first read only through the quotation in [Mak23]; its Corollary 1 has since been checked against the paper. No prior statement like Theorem D was found. The nearest are Kirby's results on countable exponentially closed fields and Zilber's pseudo-exponential fields, which are different in kind.
- *Chapter 14.* By the chapter's rubric, Theorem C is probably routine: its valuation count is the method of [BSZ17, §7, proof of Th. 33], as for the project's Laurent hull theorem. Theorem E is listed there, second among the results not found and not routine, after two blind referees scored it the same way: a complete classification, whose proof has steps not found in the sources (the shape list, the two-sided pinch, the pencil analysis). Theorem C(iii) stays routine; it is Theorem E's class (P). Theorem D is routine in method but broad in reach. Theorem F with its key step Theorem NF is listed third, below Theorem E. Its weakest point is that the specialist literature on scrolls was not read: a textbook account (Reid's *Chapters on algebraic surfaces*, Ch. 2) prints every ingredient of the bundle proof but not the three shapes, and the function-field analogues of Freiman's $3k-4$ theorem give only a containment of these spaces in Riemann–Roch spaces.

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